

Note 14.7 Extreme Values and Saddle Points

Derivative Tests for Local Extreme Values

Definition of Local Extrema(Relative Extrema)

For $f(a, b)$,

- Local maximum if $f(a, b) \geq f(x, y)$ nearby
- Local minimum if $f(a, b) \leq f(x, y)$ nearby

Theorem 10--First Derivative Test for Local Extreme Values

If $f(a, b)$ is local maximum or minimum and the first partial derivatives exist, then

$$f_x(a, b) = f_y(a, b) = 0$$

It is sufficient but not necessary.

Definition of Critical Point

The points where both partial derivatives are zero or both partial derivatives do not exist.

Not that not every critical point is correspond to a local extremum.(similar to single-variable function)

Definition of Saddle Point

The critical point that exists both $f(x, y) > f(a, b)$ and $f(x, y) < f(a, b)$ nearby. That is, the critical point that is neither local maximum nor minimum.

Theorem 11--Second Derivative Test for Local Extreme Values

	f_{xx}	$f_{xx}f_{yy} - f_{xy}^2$ ($\begin{vmatrix} < br > f_{xx} & f_{xy} \\ < br > f_{xy} & f_{yy} \\ < br > \end{vmatrix}$)
local maximum	< 0	> 0

	f_{xx}	$f_{xx}f_{yy} - f_{xy}^2$ $($ $\begin{vmatrix} < br > f_{xx} & f_{xy} \\ < br > f_{xy} & f_{yy} \\ < br > \end{vmatrix}$ $)$
local minimum	> 0	> 0
saddle point	$\backslash$	< 0
inconclusive	$\backslash$	$= 0$

Absolute Maxima and Minima on Closed Bounded Regions

- Find the local extrema of interior points.
- Find maximum and minimum on each boundary side.
- Compare and obtain the answer.

Exercises

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